

An Analysis and Simulation of Defending Tracking Control Neural Network for Known Affine Discrete-Time Nonlinear Systems Against Adversarial Attacks

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1 Abstract

The simulation of a modified tracking control neural network for known affine discrete-time nonlinear systems_[1] will be ported to C++ and optimized. A previous simulation_[1] of this system was designed in matlab. C++ gives lower level control to the programmer, allowing for more granular and programmatic optimizations to be performed. The modification to the tracking control network will to simulate adversarial attacks. For this purpose, a close analog has been chosen to represent a plausible attack scenario. Based on a previous analysis of the behavior of common adaptive cruise control systems when falsified speed data is injected into the controller area network bus_[1], a modified system equation that seeks to minimize the impact of this style of adversarial attack will be designed and simulated.

2 Introduction

A Discrete-Time Tracking Neural Network

System Model

Definition 1. *Affine-In-The-Inputs Discrete-Time Nonlinear System_[1]*

$$\underbrace{x(k+1)}_{\substack{\text{next} \\ \text{system-state}}} = \underbrace{x(k)}_{\substack{\text{current} \\ \text{system-state}}} + \underbrace{\tau}_{\substack{\text{discrete-time} \\ \text{step}}} \underbrace{\left[\underbrace{f(x(k))}_{\substack{\text{state} \\ \text{dynamics}}} + \underbrace{g(x(k))}_{\substack{\text{coupling} \\ \text{dynamics}}} \underbrace{u(k)}_{\substack{\text{control} \\ \text{input}}} \right]}_{\substack{\text{Rate of Change}}} \quad (1)$$

where:

Affine-In-The-Inputs: *The system dynamics are described by the sum of the linear function of the input and the nonlinear function of the state. This property simplifies the analysis and control design.*

Discrete-Time: *The system state is updated at specific, uniform intervals of time.*

x : *System state.*

f : *Nonlinear function that defines the system's internal state dynamics.*

g : *Nonlinear function that defines the effect of the control input on the state dynamics.*

u : *The control input applied to the system.*

τ : *Discrete-time step.*

Control Function

For the purpose of finding a solution to the tracking control problem, the control function $u(k)$ has been designed [1]:

$$u(k) = \begin{cases} g(x(k))^{-1}(-f(x(k)) + \Delta r(k) - \beta \text{sgn}(x(k) - r(k))) & \text{if } g(x(k)) \neq 0 \\ 0 & \text{if } g(x(k)) = 0 \end{cases} \quad (2)$$

such that, given a reference trajectory $r(k)$, the system (1) seeks to minimize the closed-loop error dynamics:

$$x(k+1) = x(k) - \beta \text{sgn}(x(k) - r(k)) + \Delta r(k) \quad (3)$$

where:

$$\text{sgn}(x(k) - r(k)) = \begin{cases} 1 & \text{if } x(k) > r(k) \\ 0 & \text{if } x(k) = r(k) \\ -1 & \text{if } x(k) < r(k) \end{cases} \quad (4)$$

and

$$\Delta r(k) = r(k+1) - r(k) \quad (5)$$

Intrusion Detection Systems and the Improved Resilience of Adaptive Cruise Control Systems to Spoofing Attacks

3 Adversarial Attacks

Attacks Against a Tracking Control Neural Network

The general form of the discrete-time affine-in-the-inputs nonlinear system, can be given by:

Definition 2.

$$x(k) = F(x(k), r(k)) \quad (6)$$

where:

$\mathbf{x}$: System state.

$\mathbf{F}$: Nonlinear function that defines the system's internal state dynamics.

$\mathbf{r}$: Reference trajectory that the system will attempt to follow.

4 Adversarial Attacks

Attacks Against a Tracking Control Neural Network

The general form of the discrete-time affine-in-the-inputs nonlinear system, can be given by:

Definition 3.

$$x(k) = F(x(k), r(k)) \quad (7)$$

where:

$\mathbf{x}$: System state.

$\mathbf{F}$: Nonlinear function that defines the system's internal state dynamics.

$\mathbf{r}$: Reference trajectory that the system will attempt to follow.

References

- [1] P. Tymoshchuk, "A tracking control neural network for the known affine in the inputs discrete-time nonlinear systems," 2024. University of North Texas.